

SIMATS ENGINEERING
ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.*
(BL4)

Dave's Taxonomy: Articulation (Level 4)

Question Count: 5

Duration :60 Minutes

Total Marks: 40

Code of Conduct

I Kanmani L/192511414 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student: _____

Assessment Tasks

Industry Problem Solving Task

- Retail Data Optimization Problem** - A retail chain operates both physical stores and an online platform. It generates large volumes of sales, inventory, and customer behavior data. The company wants real-time inventory tracking and demand forecasting. It also aims to deliver personalized marketing recommendations to customers. Design a big data architecture to support these requirements. Explain the tools, technologies, and analytics models you would use.
- Social Media Fraud Detection Problem** - A social media platform processes billions of user interactions daily. These include posts, comments, likes, and shares. The platform wants to detect fake accounts and misinformation quickly. Real-time or near real-time processing is required. Design a scalable big data solution for this use case. Explain the role of machine learning and stream processing frameworks.
- Government Data Platform Problem** A government agency is building a national data platform. It combines census, economic, and geospatial datasets. Different departments need access to shared and integrated data. The system must ensure interoperability across multiple agencies. Strict privacy and security requirements must be followed. Propose a governance and data management strategy.
- | | | | |
|---|--------------|------------------|----------------|
| Banking | Fraud | Detection | Problem |
| A bank processes millions of financial transactions every minute. It needs to detect fraudulent activities in real time. The system must analyze historical and streaming transaction data. Low latency and high accuracy are critical requirements. Design a big data pipeline for fraud detection. Explain the technologies and algorithms you would implement. | | | |
- | | | | |
|--|--------------|----------------|----------------|
| Agriculture | Smart | Farming | Problem |
| A smart farming system collects soil, weather, and crop data. Sensors and drones provide real-time field information. Farmers want insights to improve yield and resource usage. Data is generated from multiple distributed sources. Design a big data solution for precision agriculture. Explain how analytics can support decision-making. | | | |

Industry Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Industry Context	20%	Demonstrates strong understanding of real-world problem context	Good understanding	Basic understanding	Weak understanding
Application of Big Data Concepts	25%	Applies highly relevant big data ideas and tools	Mostly appropriate application	Partial application	Weak application
Solution Design	25%	Solution is practical, scalable, and well-reasoned	Reasonable solution	Basic solution with gaps	Unclear solution
Security / Risk Awareness	15%	Strong awareness of security and operational risks	Reasonable awareness	Limited awareness	Poor awareness
Clarity and Professionalism	15%	Well-structured and professional response	Generally clear	Moderately clear	Poorly presented

Total Marks Awarded:

1. Retail Data Optimization Problem - A retail chain operates both physical stores and an online platform. It generates large volumes of sales, inventory, and customer behavior data. The company wants real-time inventory tracking and demand forecasting. It also aims to deliver personalized marketing recommendations to customers. Design a big data architecture to support these requirements. Explain the tools, technologies, and analytics models you would use.

Answer:

A retail chain generates large volumes of sales, inventory, online transaction, and customer behavior data from both physical stores and online platforms. A big data architecture can integrate all this data and provide real-time inventory tracking, demand forecasting, and personalized marketing.

Data Collection

Data is collected from different sources such as POS systems, websites, mobile applications, warehouses, and RFID/IoT sensors. POS systems provide sales information, while online platforms provide customer searches, clicks, purchases, and browsing behavior.

- Apache Kafka is used for real-time data streaming.
- APIs connect online platforms and other business applications.
- RFID/IoT sensors help track the movement and availability of products.

Data Storage and Processing

The collected data is stored in a scalable Data Lake using technologies such as HDFS, Amazon S3, or Azure Data Lake. Both historical and real-time data can be stored for further analysis.

Apache Spark acts as the main processing engine. It can process large amounts of data in both batch and real-time modes. Spark performs data cleaning, transformation, integration, and aggregation before sending the data to analytics models.

Real-Time Inventory Tracking

The system continuously monitors stock levels across physical stores, warehouses, and the online platform. When a product is sold, the inventory information can be updated immediately.

- Detects low-stock and fast-moving products.
- Generates automatic reorder alerts.
- Helps transfer stock between stores when required.
- Reduces the chances of stock-outs and overstocking.

Demand Forecasting

Historical sales data is analyzed to predict future product demand. Machine-learning models such as ARIMA, XGBoost, Random Forest, or LSTM can be used.

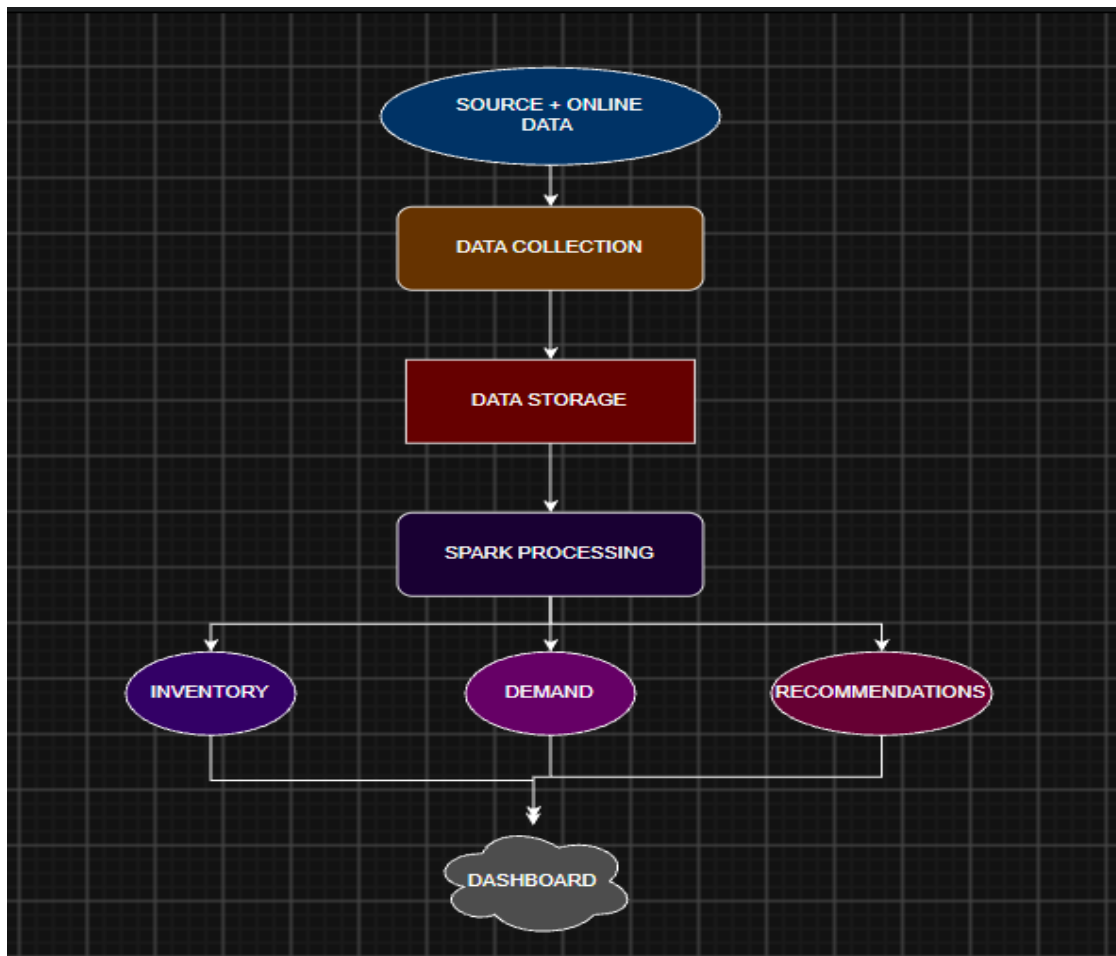
Visualization and Decision Making

The processed information can be displayed through Power BI or Tableau dashboards. Managers can monitor:

- Current inventory levels
- Sales performance
- Demand forecasts
- Low-stock alerts

Benefits:

The proposed architecture provides real-time inventory visibility, better demand prediction, and personalized customer experiences. It reduces inventory costs, prevents stock shortages, improves marketing effectiveness, and helps the retail chain make faster decisions using its large-scale data.



2. Social Media Fraud Detection Problem - A social media platform processes billions of user interactions daily. These include posts, comments, likes, and shares. The platform wants to detect fake accounts and misinformation quickly. Real-time or near real-time processing is required. Design a scalable big data solution for this use case. Explain the role of machine learning and stream processing frameworks.

Answer:

A social media platform generates billions of posts, comments, likes, shares, and user interactions every day. To detect fake accounts and misinformation quickly, the platform needs a scalable architecture that can process data in real time and use machine learning to identify suspicious behavior.

Data Collection:

Data is continuously collected from posts, comments, likes, shares, login activity, and user profiles.

- Apache Kafka can collect and distribute high-volume real-time events.
- User information and interaction patterns can be stored for both real-time and historical analysis.
- A Data Lake/HDFS or cloud storage can store historical data for training machine-learning models.

Stream Processing:

Since billions of interactions are generated continuously, batch processing alone is not sufficient. Apache Spark Streaming or Apache Flink can process incoming events with low delay.

For example, if an account suddenly creates hundreds of posts and sends thousands of friend requests within a short period, the stream-processing system can identify this unusual activity immediately.

Machine Learning for Fake Account Detection:

Machine-learning models analyze user behavior and identify patterns associated with fake accounts.

Important features include:

- Account age and profile information
- Number of followers and following
- Posting frequency
- Login patterns
- Number of likes, shares, and comments
- Unusual or repeated interactions

Misinformation Detection

Machine learning and Natural Language Processing (NLP) can analyze the text of posts and comments to identify potentially false or misleading information.

- Text classification models can classify posts as reliable or suspicious.
- NLP techniques can analyze keywords, sentiment, writing patterns, and repeated claims.
- More advanced models such as BERT/Transformer models can understand the context of posts.
- Suspicious content can be sent for fact-checking or human review before further action.

Real-Time Detection

When a suspicious event is detected, the system can immediately generate an alert.

This enables the platform to temporarily restrict suspicious accounts, flag misinformation, or send the content for manual verification.

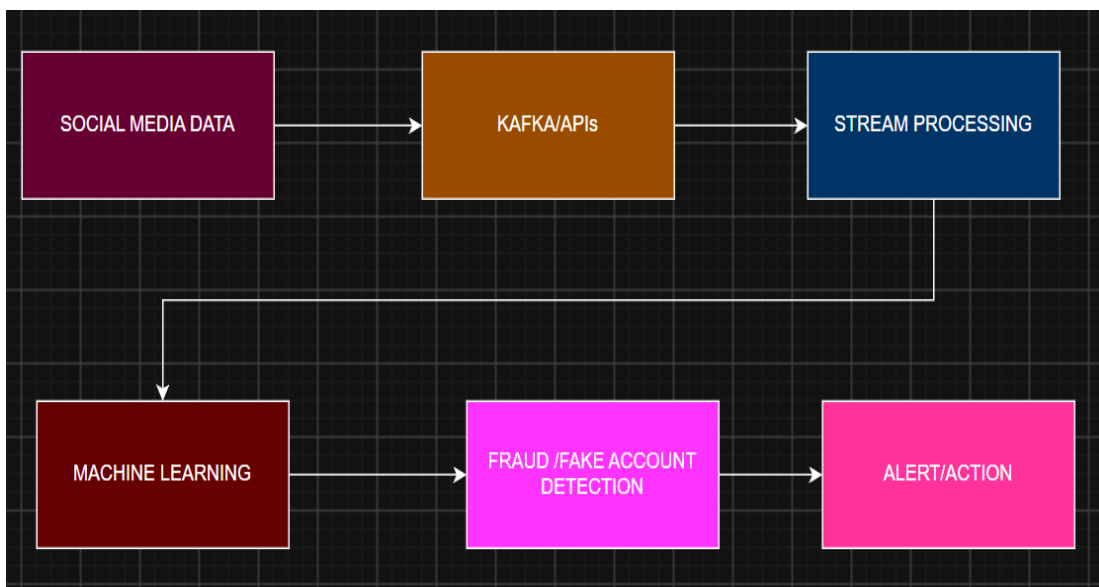
Scalability:

The system should use a distributed architecture so that processing can be performed across multiple machines.

- Kafka handles large volumes of incoming events.
- Spark/Flink distributes stream processing across multiple nodes.
- Cloud infrastructure can automatically add resources when traffic increases.
- Distributed storage such as HDFS or cloud data lakes stores large historical datasets.

Benefits:

- Detects fake accounts and misinformation with low latency.
- Handles billions of social media events.
- Machine learning improves detection accuracy over time.



3. Government Data Platform Problem A government agency is building a national data platform. It combines census, economic, and geospatial datasets. Different departments need access to shared and integrated data. The system must ensure interoperability across multiple agencies. Strict privacy and security requirements must be followed. Propose a governance and data management strategy.

Answer:

A national government data platform combines census, economic, and geospatial data from multiple departments. Since different agencies need to share and use the same information, the platform should provide data interoperability, strong governance, privacy, and security.

A central Data Governance Committee can define policies for how government data is collected, stored, shared, and used.

- Define data ownership and responsibilities for each department.
- Establish common rules for data quality, privacy, security, and access.
- Maintain a data catalog containing information about available datasets, owners, formats, and usage restrictions.
- Regularly audit data usage and compliance.

Data Integration and Interoperability

Different departments may use different databases and data formats. An integration layer can standardize and combine this information.

- APIs and ETL/ELT tools can transfer data between departments.
- Use common data formats, schemas, metadata, and naming standards.
- Geospatial data can follow standard geographic formats and services.
- A common data model allows departments to understand and use shared datasets consistently.

Data Quality Management

The platform should ensure that government data is accurate, complete, consistent, and up to date.

Data quality checks can identify:

- Missing values
- Duplicate records
- Incorrect formats
- Conflicting information between departments

Data validation and regular cleansing should be performed before important datasets are shared.

Privacy and Security

Since census and economic datasets may contain sensitive information, strong security controls are required.

- Use encryption for data at rest and during transmission.

- Apply Role-Based Access Control (RBAC) so users only access the data required for their work.
- Use authentication and multi-factor authentication for government employees.
- Sensitive information should be anonymized or pseudonymized before sharing whenever possible.
- Maintain audit logs to track who accessed or modified data.

Data Storage and Management

A centralized data lake or hybrid cloud platform can store datasets from different departments. Frequently used structured data can also be maintained in data warehouses for reporting and analytics.

Metadata and data lineage should be maintained so that departments can identify where data came from, how it was processed, and when it was updated.

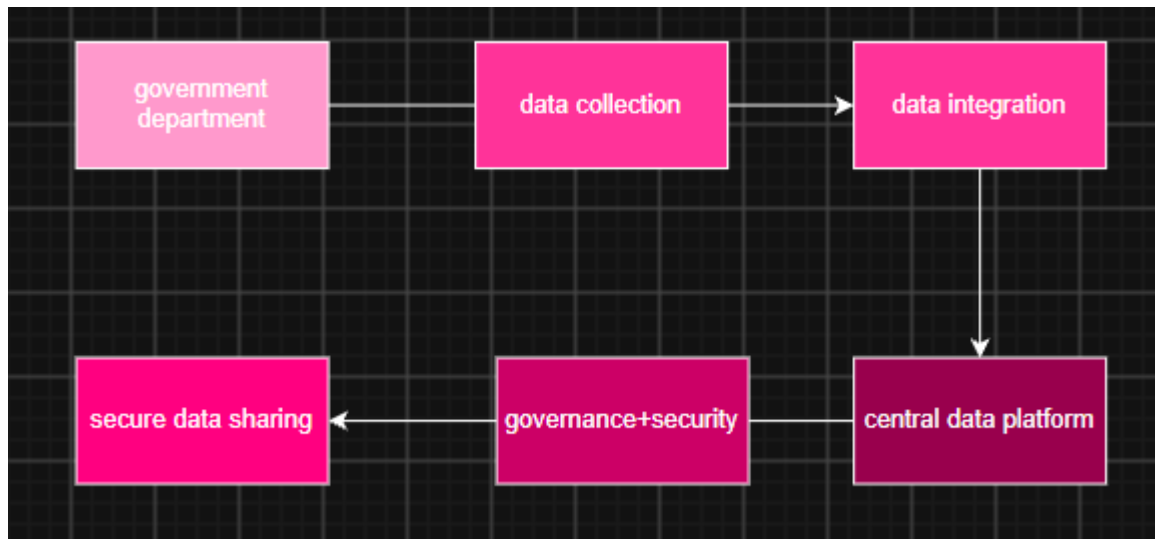
Access and Sharing

Departments can access approved datasets through a secure data portal or APIs. Access should be based on the user's role and the sensitivity of the data.

For example, a department may access aggregated census statistics, while personally identifiable information remains restricted to authorized users.

Benefits

- Provides interoperability between government departments.
- Creates a single and trusted source of shared data.
- Improves data quality and reduces duplication.



4. Banking Fraud Detection Problem

A bank processes millions of financial transactions every minute. It needs to detect fraudulent activities in real time. The system must analyze historical and streaming transaction data. Low latency and high accuracy are critical requirements. Design a big data pipeline for fraud detection. Explain the technologies and algorithms you would implement.

Answer:

A bank processes millions of transactions such as ATM withdrawals, online payments, card transactions, and fund transfers every minute. Since fraudulent transactions must be detected immediately, the system should combine real-time stream processing with historical data analysis.

Data Collection

Transaction data is continuously collected from ATMs, credit/debit cards, online banking, mobile banking, and payment gateways.

- Apache Kafka can collect millions of transaction events in real time.
- Historical transaction records are stored in a Data Lake/HDFS or cloud storage.
- Important information includes transaction amount, location, time, merchant, device, and customer behavior.

Stream Processing

Since fraud detection requires very low latency, Apache Spark Streaming or Apache Flink can process transactions as they arrive.

The system can calculate features such as:

- Number of transactions within a short time
- Unusual transaction amount
- Sudden change in location
- New or unknown device
- Repeated failed transactions

Machine Learning Algorithms

The system can use both supervised and unsupervised learning.

- Random Forest / XGBoost: Classify transactions as legitimate or fraudulent using historical labelled data.
- Logistic Regression: Provides a simple and fast fraud classification model.
- Isolation Forest: Detects unusual transactions when labelled fraud data is limited.
- Autoencoders: Identify transactions that significantly differ from normal customer behavior.

The models can generate a fraud probability score for every transaction. Transactions above a predefined threshold can be flagged for further investigation.

Real-Time Response

The system should respond immediately after detecting suspicious activity.

- Low-risk: Approve the transaction.
- Medium-risk: Request additional authentication such as OTP.
- High-risk: Temporarily block the transaction and generate an alert.

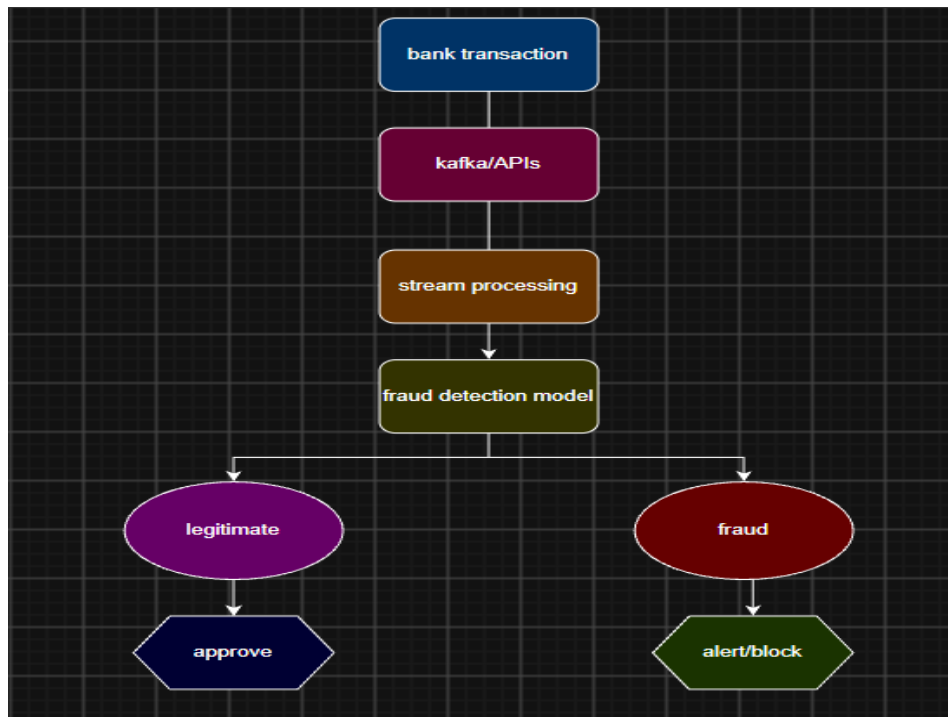
This reduces financial losses while avoiding unnecessary blocking of genuine customers

Low Latency and Accuracy

To achieve low latency, transactions should be processed through Kafka and Spark/Flink without waiting for batch processing. In-memory processing and distributed computing allow the system to handle millions of transactions simultaneously.

Benefits

- Detects fraudulent transactions in near real time.
- Handles millions of transactions using distributed processing.
- Combines historical and streaming data for better accuracy.



5. Agriculture Smart Farming Problem

A smart farming system collects soil, weather, and crop data. Sensors and drones provide real-time field information. Farmers want insights to improve yield and resource usage. Data is generated from multiple distributed sources. Design a big data solution for precision agriculture. Explain how analytics can support decision-making.

Answer:

A smart farming system collects large amounts of soil, weather, crop, and field data using sensors, drones, satellites, and other distributed sources. A big data solution can combine this information to help farmers improve crop yield, irrigation, fertilizer usage, and overall resource management.

Data Collection

Data is collected continuously from different sources across the farm.

- Soil sensors: moisture, temperature, pH, and nutrient levels.
- Weather stations: rainfall, humidity, temperature, and wind speed.
- Drones and satellites: crop images and vegetation information.
- Farm equipment: information about irrigation, fertilizers, and machinery.

IoT gateways and MQTT/Kafka can collect and transmit this distributed data to the processing system.

Data Storage and Processing

The collected data can be stored in a cloud-based Data Lake, HDFS, or Amazon S3. This allows large amounts of sensor readings and images to be stored efficiently.

Apache Spark can process the data and combine information from different sources. It can perform data cleaning, filtering, aggregation, and real-time analysis.

Analytics for Precision Agriculture

Machine-learning models can analyze historical and real-time farm data to support better decisions.

- Yield Prediction: Regression models, Random Forest, or XGBoost can predict crop yield using soil, weather, and crop conditions.
- Crop Health: Drone and satellite images can be analyzed using image processing or CNN models to identify unhealthy crops, pests, or diseases.
- Irrigation Prediction: Soil moisture and weather data can determine when and how much water is required.
- Disease Detection: Image-based ML models can detect early signs of crop diseases.

Resource Optimization

Analytics helps farmers use resources only where they are required.

For example, if sensors show that a particular field area has sufficient soil moisture, irrigation can be reduced in that area. Similarly, fertilizer recommendations can be generated based on soil nutrient levels and crop requirements.

This reduces water, fertilizer, and energy consumption while maintaining or improving crop yield.

Real-Time Alerts

The system can generate alerts when abnormal conditions are detected.

- Low soil moisture → Irrigation alert
- Extreme temperature → Weather alert
- Possible crop disease → Disease alert
- Low nutrient level → Fertilizer recommendation

Benefits

- Improves crop yield and quality.
- Reduces water, fertilizer, and energy usage.
- Detects crop diseases at an early stage.

